ECONOMETRICS - GROUP H ASSIGNMENTS

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*1. LOAD THE DATASET AND EXAMINE THE TIME SERIES OF **INF** AND **UNEM** FROM 1948 TO 2003.*

VISUALIZE TRENDS, IDENTIFYING MAJOR MACROECONOMIC EVENTS (E.G., STAGFLATION OF THE 1970S, DISINFLATION POST-1980). COMPUTE AND PLOT DESCRIPTIVE STATISTICS, LAGS, AND FIRST DIFFERENCES.

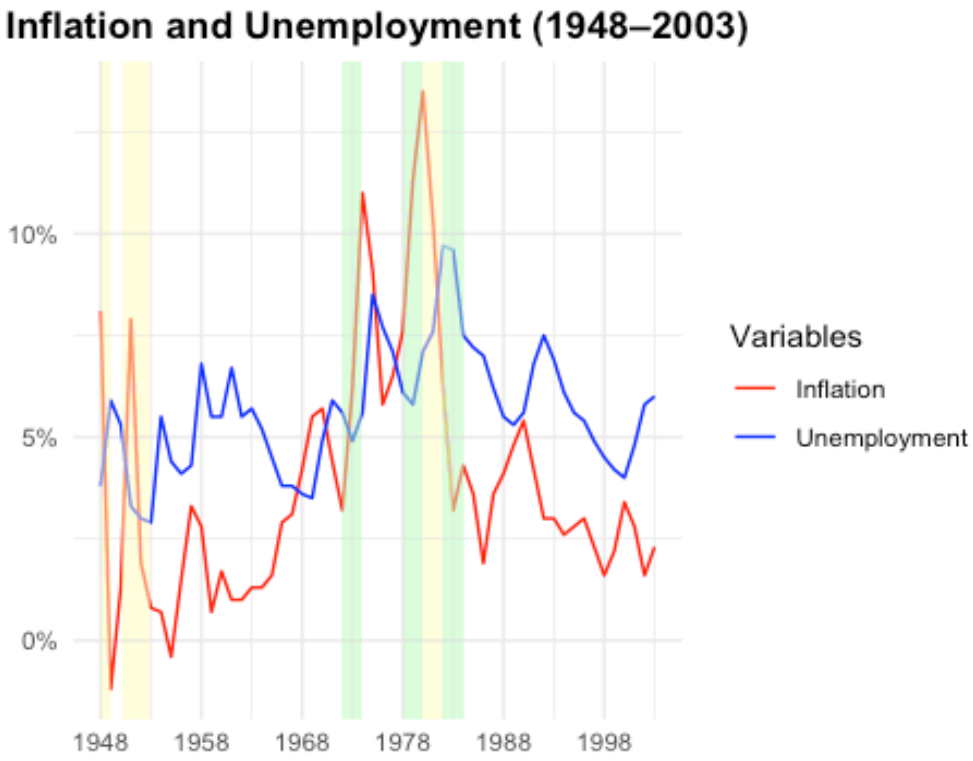
```
#Download the dataset "phillips.xlsx"

library(readxl)

dati <- read_excel("phillips.xlsx", col_types = c("numeric", "numeric", "numeric",
 "numeric", "numeric", "numeric", "numeric", "numeric"))

# Plot with ggplot2
library(ggplot2)
library(scales)

ggplot(dati, aes(x = year)) +
  geom_line(aes(y = inf, color = "Inflation")) +
  geom_line(aes(y = unem, color = "Unemployment")) +
  scale_color_manual(values = c("Inflation" = "red", "Unemployment" = "blue")) +
  scale_x_continuous(breaks = seq(1948, 2003, by = 10)) +
  scale_y_continuous(labels = label_percent(scale = 1)) +
  labs(title = "Inflation and Unemployment (1948–2003)",
       x = " ", y = " ", color = "Variables") +
  geom_rect(xmin = 1980, xmax = 1983, ymin = -Inf, ymax = Inf,
            fill = "wheat1", alpha = 0.01) +
  geom_rect(xmin = 1948, xmax = 1949, ymin = -Inf, ymax = Inf,
            fill = "wheat1", alpha = 0.01) +
  geom_rect(xmin = 1950, xmax = 1953, ymin = -Inf, ymax = Inf,
            fill = "wheat1", alpha = 0.01) +
  geom_rect(xmin = 1972, xmax = 1974, ymin = -Inf, ymax = Inf,
            fill = "darkseagreen1", alpha = 0.01) +
  geom_rect(xmin = 1978, xmax = 1980, ymin = -Inf, ymax = Inf,
            fill = "darkseagreen1", alpha = 0.01) +
  geom_rect(xmin = 1982, xmax = 1984, ymin = -Inf, ymax = Inf,
            fill = "darkseagreen1", alpha = 0.01) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```




```
#1950: inflation spike due to the Korean War
#1970: stagflation: with high levels of unemployment and inflation
#1973 and 1979: oil crisis
#1980: unemployment peak (this may be a consequence of the restrictive monetary po
licies adopted to combat inflation)

#descriptive statistics for the variable "inf"
library(psych)

describe(dati$inf)

##    vars  n mean   sd median trimmed  mad  min  max range skew kurtosis   se
## X1    1 56 3.88 3.04   3.05    3.52 2.15 -1.2 13.5  14.7 1.15     1.02 0.41

#descriptive statistics for the variable "unem"
describe(dati$unem)

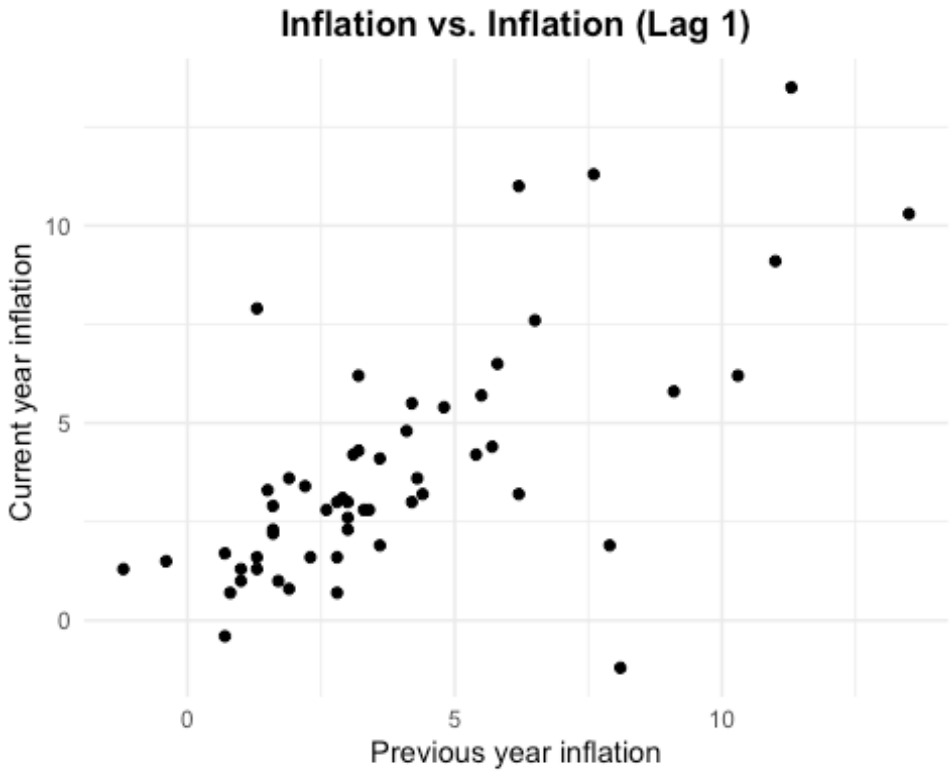
##    vars  n mean   sd median trimmed  mad min max range skew kurtosis  se
## X1    1 56 5.63 1.51   5.55    5.57 1.63 2.9 9.7   6.8 0.49     0.12 0.2

#lags
library(dplyr)

# Calculate the lag 1 (previous period value) for inflation (inf) and unemployment
 (unem)
```

```r
dati <- dati %>% mutate(inf_lag1 = lag(inf, n = 1),
                        unem_lag1 = lag(unem, n = 1))

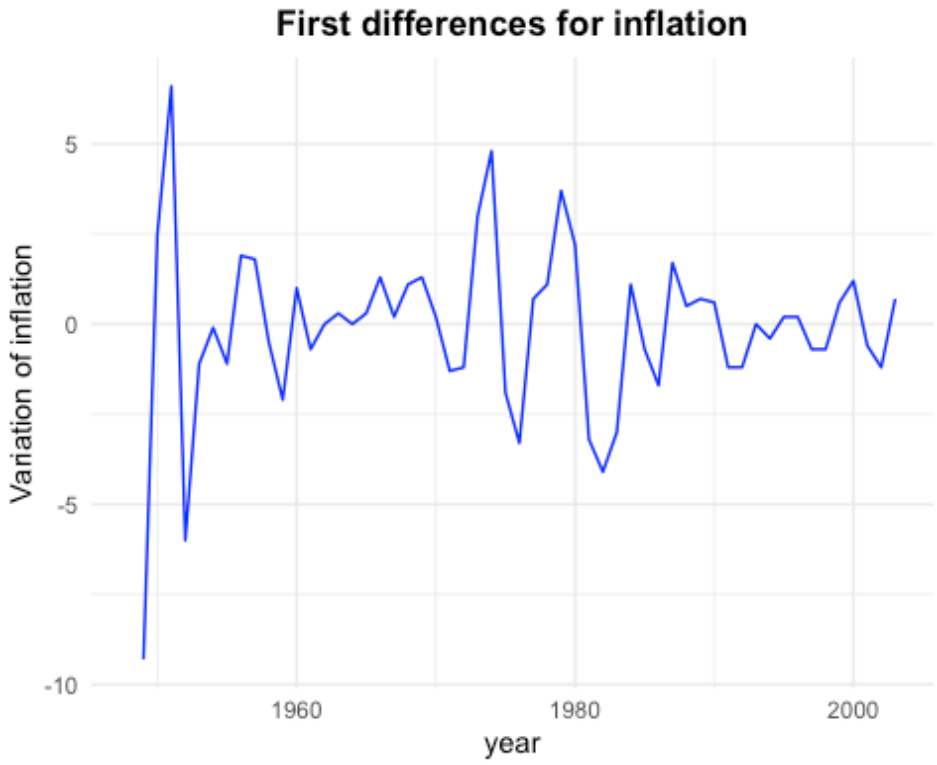
# Visualize the original series vs. lag 1
# This helps you see the relationship between the current and past values
ggplot(dati, aes(x = inf_lag1, y = inf)) +
  geom_point() +
  labs(title = "Inflation vs. Inflation (Lag 1)",
       x = "Previous year inflation",
       y = "Current year inflation ") +
  theme_minimal()+
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



This lag plot suggests that inflation tends to persist over time. One year's inflation is an indicator (though not perfect) of the following year's inflation. However, the scatter of the points indicates that other factors play a role in determining the level of inflation.

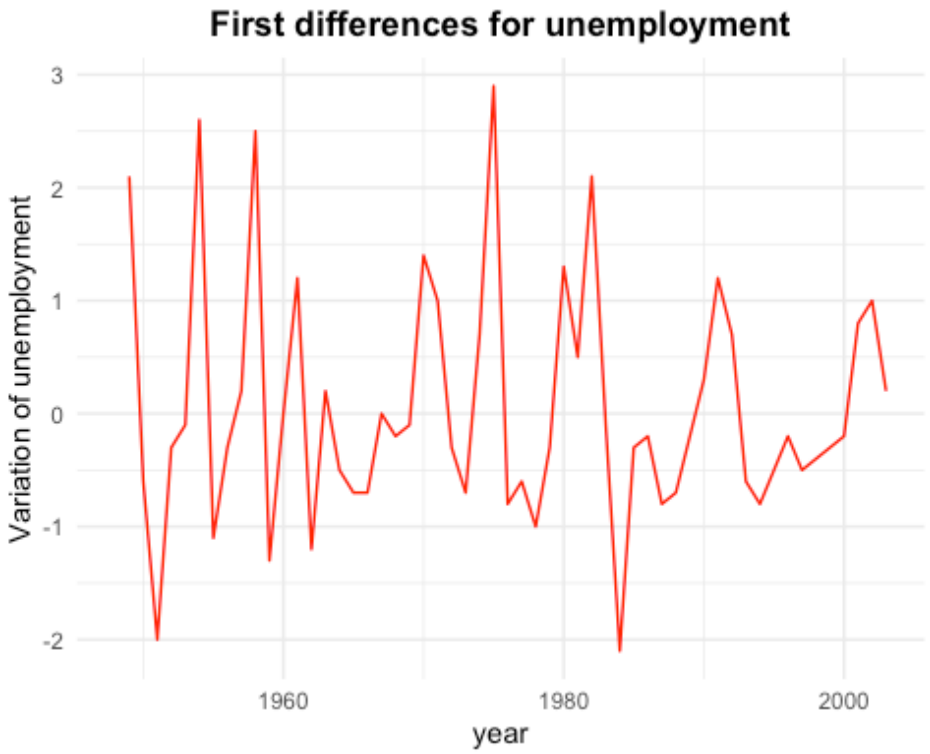
```r
#first differences
dati <- dati %>%
  mutate(
    inf_diff = inf - lag(inf, n = 1),
    unem_diff = unem - lag(unem, n = 1)
  )
#inflation
ggplot(dati, aes(x = year)) +
```

```
  geom_line(aes(y = inf_diff), color = "blue") +
  labs(title = "First differences for inflation",
       y = "Variation of inflation") +
  theme_minimal()+
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



This graph shows the annual change in the inflation rate. Large oscillations indicate high inflation volatility, with peaks above zero signaling increases compared to the previous year, and valleys below zero indicating decreases compared to the previous year (disinflation). Periods near zero show stability.

```
ggplot(dati, aes(x = year)) +
  geom_line(aes(y = unem_diff), color = "red") +
  labs(title = "First differences for unemployment",
       y = "Variation of unemployment") +
  theme_minimal()+
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



This graph shows the annual change in the unemployment rate. Frequent oscillations indicate that the unemployment rate changes year to year, with peaks above zero signaling increases and valleys below zero indicating decreases. Periods near zero show stability.

2. ESTIMATE THE CLASSICAL PHILLIPS CURVE MODEL:

$$inf_t = \beta_0 + \beta_1 \cdot unem_t + u_t$$

EXPLAIN THE EXPECTED SIGN OF $\beta 1$ BASED ON ECONOMIC THEORY. WHAT DOES THE SLOPE TELL US ABOUT THE INFLATION-UNEMPLOYMENT TRADEOFF?

```
modello_phillips <- lm(inf ~ unem, data = dati)
summary(modello_phillips)

##
## Call:
## lm(formula = inf ~ unem, data = dati)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.2176 -1.7812 -0.6659  1.1473  8.8795
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)   1.0536     1.5480   0.681   0.4990
## unem          0.5024     0.2656   1.892   0.0639 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.972 on 54 degrees of freedom
## Multiple R-squared:  0.06215,    Adjusted R-squared:  0.04479
## F-statistic: 3.579 on 1 and 54 DF,  p-value: 0.06389
```

The estimated coefficient for the unemployment rate (β1=0.5024) is positive. A positive β1 suggests that, during the period analyzed (1948-2003), higher unemployment appears to be associated with higher inflation. This is the opposite of what would be expected according to the short-run Phillips Curve, which posits a trade-off between unemployment and inflation: an increase in unemployment should lead to a decrease in inflation, and vice versa.

The p-value associated with this estimated coefficient is 0.0639. This means that at a 5% level of significance, the positive relationship is not statistically significant. Therefore, there isn't strong statistical evidence to conclude that a positive relationship exists between unemployment and inflation for the period 1948-2003 based on this model.

The Phillips Curve suggests that, in the long run, there isn't a stable trade-off between inflation and unemployment. According to this very simple model, inflation expectations play a crucial role. If inflation expectations increase, both inflation and unemployment could rise simultaneously. The period considered for the analysis is characterized by major macroeconomic shocks (as highlighted in point 1) that influenced inflation expectations.

Another factor that influences the relationship between the two variables is the negative supply shock (like an increase in commodity prices), which may have led to a rise in both inflation and unemployment—as occurred in the 1970s during a period of stagflation. Finally, the time period analyzed is quite broad, so by considering sub-periods and including additional variables in the model, the estimated coefficient β_1 could differ.

3. AUGMENT YOUR MODEL TO INCLUDE DYNAMICS:

$$inf_t = \beta_0 + \beta_1 \cdot unem_t + \beta_2 \cdot inf_{t-1} + u_t$$

DISCUSS THE ROLE OF INFLATION EXPECTATIONS AND INERTIA IN THE MODERN PHILLIPS CURVE.

```
#Computation of the model
model <- lm(inf ~ unem + inf_lag1, data = dati)
summary(model)

##
## Call:
## lm(formula = inf ~ unem + inf_lag1, data = dati)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.0239 -0.9088 -0.1475  0.7533  5.5013
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.2100     1.2100   1.826   0.0735 .
## unem         -0.2241     0.2389  -0.938   0.3525
## inf_lag1      0.7329     0.1172   6.251  7.7e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.221 on 52 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.477,  Adjusted R-squared:  0.4568
## F-statistic: 23.71 on 2 and 52 DF,  p-value: 4.806e-08
```

The coefficient β_1 is −0.2241, which aligns with the theory behind the Phillips Curve: the inverse relationship between unemployment and inflation. However, the associated level of statistical significance is low, meaning we cannot confidently claim that unemployment has a negative impact on inflation.

In contrast, the coefficient β_2, which captures inflation inertia (the influence of past inflation on current inflation), is quite large (0.7329) and highly significant. This provides strong evidence of a positive and persistent effect of past inflation on current inflation, suggesting that inflation tends to persist over time.

These findings highlight the role of inflation expectations, which appear to be adaptive: economic agents base their current decisions on past inflation, expecting future inflation to resemble recent trends. In this way, inflation expectations influence actual inflation outcomes, as reflected in the strong effect of lagged inflation.

4. SPLIT THE SAMPLE INTO TWO PERIODS: 1948–1979 (PRE-VOLCKER) AND 1980–2003 (POST-VOLCKER DISINFLATION):

- *Estimate the model separately for each period.*
- *Perform a Chow test to determine whether the relationship between inflation and unemployment changed structurally after 1980.*
- *Discuss economic reasons for the shift (e.g., credibility of monetary policy, globalization, labor market flexibility).*

```
# Pre-Volcker (1948-1979)
df_pre_vol <- dati[dati$year >= 1948 & dati$year <= 1979, ]

# Post-Volcker  (1980-2003)
df_post_vol <- dati[dati$year >= 1980 & dati$year <= 2003, ]
```

Estimation of the model for each period.

```r
# Pre-Volcker period
model_pre_vol <- lm(inf ~ unem + inf_lag1, data = df_pre_vol)
summary(model_pre_vol)
```

```
##
## Call:
## lm(formula = inf ~ unem + inf_lag1, data = df_pre_vol)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.6349 -1.1192 -0.2222  0.9306  5.7242
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.7088     2.0282   0.843  0.40663
## unem         -0.0466     0.4212  -0.111  0.91270
## inf_lag1      0.6174     0.1874   3.295  0.00267 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.767 on 28 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.3243, Adjusted R-squared:  0.2761
## F-statistic: 6.721 on 2 and 28 DF,  p-value: 0.004132
```

The estimated coefficient for unemployment is -0.0466. While negative, aligning with the traditional Phillips Curve theory, it is not statistically significant (p-value = 0.91270). This suggests that, in the pre-Volcker era, a change in unemployment did not have a statistically robust impact on inflation, once past inflation is accounted for. The Adjusted R-squared is 0.2761. This means that about 27.61% of the variation in inflation during the pre-Volcker period is explained by the model.

The relationship between unemployment and inflation appears weak or statistically insignificant. However, there's strong evidence of inflationary persistence, where past inflation plays a predominant role in explaining current inflation. This finding is consistent with a period marked by oil shocks and rising inflationary expectations, which contributed to high inflation even among high unemployment (stagflation).

```r
# Post-Volcker period
model_post_vol <- lm(inf ~ unem + inf_lag1, data = df_post_vol)
summary(model_post_vol)
```

```
##
## Call:
## lm(formula = inf ~ unem + inf_lag1, data = df_post_vol)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -1.4799 -1.0277 -0.1067  0.5544  3.5483
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.2349     1.2200   2.652   0.0149 *
## unem         -0.5196     0.2263  -2.296   0.0321 *
## inf_lag1      0.9209     0.1087   8.469 3.26e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.272 on 21 degrees of freedom
## Multiple R-squared:  0.802,  Adjusted R-squared:  0.7832
## F-statistic: 42.53 on 2 and 21 DF,  p-value: 4.117e-08
```

The estimated coefficient unem is -0.5196 and is statistically significant (p-value = 0.0321). This marks a critical shift from the pre-Volcker period. The sign is negative, as expected by the theoretical Phillips Curve: a one percentage point increase in unemployment is associated with an approximate 0.52 percentage point decrease in inflation. This suggests that the trade-off relationship between inflation and unemployment re-emerged in the post-Volcker era.

The Adjusted R-squared is 0.7832. This is a very high value, indicating that about 78.32% of the variation in inflation during the post-Volcker period is explained by the model.

```
##Perform a Chow test
library(strucchange)

#Preparation to ChowTest
#We need a breakpoint, in our case is 1980
bp_idx <- which(dati$year == 1980)

#Chow Test
chow_test <- sctest(model, point = bp_idx, type = "Chow")
chow_test

##
##  M-fluctuation test
##
## data:  model
## f(efp) = 1.3142, p-value = 0.1779
```

Since the p-value (0.1779) is greater than commonly accepted significance levels, we fail to reject the null hypothesis. This means that, based on this Chow test, there is insufficient statistical evidence to conclude that the relationship between inflation, unemployment, and lagged inflation fundamentally changed structurally after 1980.

This outcome might seem to go against the conclusions given by individual coefficients changed noticeably between the two periods in the separate estimations. However, the Chow test evaluates the overall change of all coefficients jointly. It's possible that, while there were shifts in individual

relationships, the overall structure didn't change drastically enough to be statistically significant at standard confidence levels, or that the test is not powerful enough.

```
# Discuss economic reasons for the shift (e.g., credibility of monetary policy, globalization, labor market flexibility).
```

Between 1948 and 2003, the relationship between inflation and unemployment in the United States changed profoundly. Although the Chow test does not detect a clear structural break, the differences between the pre-Volcker period (1948–1979) and the post-Volcker period (1980–2003) indicate an important economic shift, which can be explained by several key economic factors.

First of all, a central role was played by the credibility of monetary policy. In the 1970s, the Federal Reserve was perceived as lacking credibility and unable to fight inflation, which led to unanchored inflation expectations: businesses and citizens expected inflation to continue or increase. With the appointment of Paul Volcker in 1979, the Fed adopted a tough stance, drastically raising interest rates even at the cost of a deep recession. This determined approach restored confidence in the central bank's ability to maintain price stability.

Secondly, globalization had a significant impact. The increase in international competition reduced the power of firms to raise prices, and the offshoring of production helped contain wage pressures. These factors contributed to reducing cost-push inflation.

In summary, greater credibility of monetary policy, globalization, and changes in the labor market explain the shift, after 1980, toward lower and more stable inflation.

5. TEST FOR MULTICOLLINEARITY (VIF), HETEROSKEDASTICITY (BREUSCH-PAGAN), AND SPECIFICATION ERRORS (RESET TEST). REPORT WHETHER YOUR MODEL ASSUMPTIONS HOLD.

```
library(car)

library(lmtest)
library(sandwich)

#test for multicollinearity (VIF)
vif(model)

##     unem inf_lag1
## 1.409761 1.409761

#compute the heteroskedasticity test (Breusch-Pagan)
bptest(model)

##
##  studentized Breusch-Pagan test
##
## data:  model
## BP = 12.189, df = 2, p-value = 0.002255
```

```r
#test for specification errors (test RESET)
resettest(model)

##
##  RESET test
##
## data:  model
## RESET = 0.05783, df1 = 2, df2 = 50, p-value = 0.9439
```

We evaluated the linear regression model against key assumptions. The Variance Inflation Factor (VIF) values were low (1.41 for both unem and inf_lag1), well below the critical threshold of 5, indicating no issues with multicollinearity.

However, the Breusch-Pagan test revealed evidence of heteroskedasticity (p-value = 0.0023), leading us to reject the null hypothesis of homoskedasticity and suggesting that the assumption of constant variance in the residuals is violated.

Conversely, the RESET test produced a p-value of 0.94, confirming that the model is correctly specified with no apparent structural misspecifications, such as omitted variables or incorrect functional form. Overall, while the model is properly specified and free from multicollinearity concerns, the presence of heteroskedasticity suggests that robust standard errors may be necessary to ensure valid statistical inference.

6. USING THE BEST-FITTING MODEL FROM YOUR EARLIER ANALYSIS, FORECAST INFLATION FOR THE NEXT 3 YEARS BASED ON HYPOTHETICAL UNEMPLOYMENT SCENARIOS (E.G., 5%, 6%, 7%). DISCUSS CONFIDENCE IN YOUR FORECAST AND LIMITATIONS.

```r
library(forecast)

#Let's transform the inflation and unemployment columns into time series.
inf_ts <- ts(dati$inf, start = c(1948, 1), frequency = 1)
unem_ts <- ts(dati$unem, start = c(1948, 1), frequency = 1)

#The best-fitting model is the Phillips Curve augmented of point 3
#This corresponds to an AR(1) model with 'unem' as an external regressor.
best_model_arima <- forecast::Arima(inf_ts, #Dependent variable
                          order = c(1,0,0), # This row defines p=1, d=0, q=0
                          xreg = unem_ts, #Exogenous variable
                          include.mean = TRUE)
summary(best_model_arima)

## Series: inf_ts
## Regression with ARIMA(1,0,0) errors
##
## Coefficients:
##          ar1  intercept     xreg
##       0.7837     7.9632  -0.7078
## s.e.  0.0852     2.0996   0.3051
```

```
##
## sigma^2 = 4.66:  log likelihood = -121.49
## AIC=250.98   AICc=251.76   BIC=259.08
##
## Training set error measures:
##                       ME     RMSE      MAE      MPE     MAPE      MASE      ACF1
## Training set -0.05872128 2.100131 1.40786 1.113029 69.23029 0.8719855 0.0392528
```

In our model ($inft = \beta 0 + \beta 1 \cdot unemt + \beta 2 \cdot inft{-}1 + u$), B2 is equal to 0.7837, so the inflation in the current period is positively influenced by inflation from the previous period. A value of 0.7837 suggests strong persistence: if inflation was 1% higher last year, it is expected to be approximately 0.78% higher this year, all else being equal.

The intercept, which is the expected level of inflation when unemployment (xreg) and past inflation (arl) are zero, is 7.9632.

The xreg coefficient indicates the effect of unemployment on inflation. The negative sign (-0.7078) is consistent with the economic theory of the Phillips Curve: an increase in unemployment is associated with a decrease in inflation (or vice versa). If unemployment rises by 1 percentage point, inflation is predicted to decrease by approximately 0.71 percentage points, all else being equal (or holding past inflation constant).

```
#Let's forecast for 3 years
forecast_horizon <- 3

#Scenario 1: Unemployment constant at 5%
future_unem_5pct <- ts(rep(0.05, forecast_horizon), # 'rep(0.05, 3)' create a vect
or with 3x 0.05
                       start = c(2004, 1),
                       frequency = 1)      # annual data
# Scenario 2: Unemployment constant at 6%
future_unem_6pct <- ts(rep(0.06, forecast_horizon), start = c(2004, 1), frequency
= 1)

# Scenario 3: Unemployment constant at 7%
future_unem_7pct <- ts(rep(0.07, forecast_horizon), start = c(2004, 1), frequency
= 1)

forecast_5pct <- forecast(best_model_arima, h = forecast_horizon, xreg = future_un
em_5pct)
forecast_6pct <- forecast(best_model_arima, h = forecast_horizon, xreg = future_un
em_6pct)
forecast_7pct <- forecast(best_model_arima, h = forecast_horizon, xreg = future_un
em_7pct)

inflation_forecasts_5pct <- as.numeric(forecast_5pct$mean)
```

```r
inflation_forecasts_6pct <- as.numeric(forecast_6pct$mean)
inflation_forecasts_7pct <- as.numeric(forecast_7pct$mean)

#The mean of a 'forecast' object contains the point estimate
forecast_table <- data.frame(
  Year = 2004:(2003 + forecast_horizon),
  Inf_5pct_Unemp = inflation_forecasts_5pct,
  Inf_6pct_Unemp = inflation_forecasts_6pct,
  Inf_7pct_Unemp = inflation_forecasts_7pct
)
forecast_table
```

```
##   Year Inf_5pct_Unemp Inf_6pct_Unemp Inf_7pct_Unemp
## 1 2004       6.817905       6.810827       6.803748
## 2 2005       7.057986       7.050908       7.043829
## 3 2006       7.246136       7.239057       7.231979
```

As can be observed from the table, as the unemployment rate increases (moving from 5% to 6% and then to 7%), the forecasted inflation rate slightly decreases. The magnitude of unemployment's effect on inflation appears to be quite modest in the forecasts. This suggests that the Phillips Curve might be relatively flat during the forecasted period, which implies that significant changes in unemployment are necessary to achieve large changes in inflation, or vice versa.

Regarding confidence in the forecast, we can consider the MASE (Mean Absolute Scaled Error) value of 0.87, which suggests an improvement compared to a naive model (a model that predicts the last observed value at time t as the forecast for time t+1)

Furthermore, the negative sign of the unemployment coefficient (-0.7078) is consistent with the theory of the Phillips Curve. This strengthens the model's credibility as a plausible representation of inflationary dynamics.

It can also be noted that the ACF1 is close to 0: a low ACF1 implies that the model has extracted the available information from the series in terms of its relationship with its immediate past and with the included exogenous variables.

Regarding limitations, it is important to emphasize that the forecasts depend on the hypothetical unemployment scenarios provided (5%, 6%, 7%). Confidence in the forecasts is, therefore, linked to confidence in the unemployment scenarios. Furthermore, the model cannot predict future exogenous shocks not captured by the included variables. An important aspect to consider, moreover, is that the model is linear and does not account for any non-linearities or regime changes in the relationship between inflation and unemployment that might emerge.

7. WHAT DO YOUR RESULTS IMPLY ABOUT THE TRADEOFF BETWEEN UNEMPLOYMENT AND INFLATION? HOW DOES THIS TRADEOFF CHANGE OVER TIME? DISCUSS THE IMPLICATIONS FOR CENTRAL BANK POLICY AND INFLATION TARGETING.

8. BASED ON YOUR FINDINGS, SHOULD MONETARY POLICY PLACE MORE EMPHASIS ON INFLATION STABILIZATION OR EMPLOYMENT? HOW RELIABLE IS THE PHILLIPS CURVE AS A TOOL FOR MODERN MACROECONOMIC MANAGEMENT?

Regarding the trade-off between inflation and unemployment in the model, our results indicate a negative correlation when considering the previous year's inflation value, although this relationship is not very pronounced. It can be stated that the Phillips Curve appears relatively flat during the analyzed period, suggesting that an increase in unemployment translates into only a slight decrease in inflation.

However, it is important to note that within the dataset's analyzed period (1948-2003), there were phases of stagflation, particularly in the 1970s, partly due to oil shocks, during which both inflation and unemployment rapidly increased. In these specific periods, the negative correlation between the two variables diminished or disappeared.

Furthermore, it is observed that inflation tends to persist over time, while unemployment is much more volatile and does not appear to be strongly linked to its value from the previous year. This difference could be attributed to the influence of adaptive expectations for inflation, which may lead to a slower recovery of confidence in the system.

A relatively flat Phillips Curve suggests that reducing inflation by increasing unemployment is either ineffective or very costly: even a sharp rise in unemployment would only lead to a slight decrease in inflation. This limits the usefulness of unemployment as a tool for controlling inflation without causing significant economic harm.

Stagflation shows that the traditional trade-off between inflation and unemployment can break down, putting central banks in a dilemma: fighting inflation may worsen unemployment, and addressing unemployment may fuel inflation. In such cases, it's essential to consider external factors that can reshape macroeconomic dynamics.

When facing major supply-side shocks, a more flexible inflation targeting approach may be appropriate. The central bank should take into account both inflation and the output gap, while keeping price stability as its main medium-term goal.

Because adaptive expectations make inflation persistent, monetary policy must be credible and proactive in anchoring expectations. A clearly communicated inflation target helps reinforce this anchor and break the wage-price spiral.

Finally, unemployment tends to be more volatile and less persistent than inflation, making it more responsive and less predictable in reaction to monetary policy. This adds complexity to managing unemployment compared to inflation.

Although the Phillips Curve remains a useful conceptual framework, its reliability as a practical tool in modern macroeconomics is ambiguous. Structural changes in the economy, globalization, and the evolution of the labor market have weakened the historical relationship between inflation and unemployment. As a result, it should be complemented with other indicators and models to gain a more comprehensive understanding of economic conditions.